

Research paper

Jahn-Teller effect in high spin d^4 and d^9 octahedral metal-complexes

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ABSTRACT

The Jahn-Teller effect, which is the geometrical effect on a molecule due to the energy lowering of spatially degenerate electronic ground states, is not determined, as is often perceived, by the shape of the highest occupied molecular orbital (HOMO), or the shape of a singly occupied molecular orbital (SOMO), but is determined rather by the difference in concentration of electron density (before distortion) between the metal and the two ligands on the z-axis, and concentration of electron density between the metal and the four ligands lying in the xy-plane. A higher concentration of electron density between the metal and the two ligands on the z-axis, than between the metal and the four ligands on the xy-plane, will lead to elongation (z-out) Jahn-Teller distortion. Accordingly, high spin, $S = 2$, d^4 octahedral metal-complexes with a d_{z^2} HOMO, as well as $S = 1/2$, d^9 octahedral metal-complexes with a $d_{x^2-y^2}$ HOMO, both exhibit elongation (z-out) Jahn-Teller distortion.

1. Introduction

The Jahn-Teller effect (JTE) can be described as the geometrical distortion of molecules and ions which are associated with certain electron configurations. This contribution focuses on the JTE observed in high spin, $S = 2$, d^4 and $S = 1/2$, d^9 octahedral transition metal complexes, and explains why an unpaired electron in the d_{z^2} HOMO orbital of a high spin d^4 octahedral transition metal compound, exhibits elongation (z-out) Jahn-Teller distortion, while an unpaired electron in the d_{z^2} HOMO orbital of a d^9 octahedral transition metal compound, exhibits compression (z-in) Jahn-Teller distortion instead. This discussion serves as a systematic approach in understanding the JTE, using two examples of molecules that have different degenerate electronic ground states.

The Jahn-Teller effect or theorem has been known since 1937, when Hermann Jahn and Edward Teller used symmetry arguments to illustrate that any nonlinear molecule with a spatially degenerate electronic ground state, will distort itself in such a way that the degenerate energy levels will split, resulting in the lowering of symmetry of the molecule, as well as in the lowering of the overall energy of the species [1]. An electronically degenerate state represents the availability of more than one degenerate orbital for an electron. See Fig. 1 for the two possible degenerate states for both high spin, $S = 2$, d^4 ($t_{2g}^3 e_g^1$) and for $S = 1/2$, d^9 ($t_{2g}^6 e_g^3$) octahedral transition metal complexes. The electronic distortion will result in a change of the electron distribution within the molecule (splitting of energy levels), as well as in a change of the spatial arrangement of the atoms in the molecule (geometrical distortion). This Jahn-Teller distortion is most commonly observed in octahedral

transition metal complexes, however, it can also be observed in tetrahedral compounds. Figure S1 in the Supporting Information shows the electron filling for the t_{2g} and e_g energy levels of octahedral $d^1 - d^{10}$ metal complexes, also indicating which complexes are susceptible to Jahn-Teller distortion. Octahedral complexes with Jahn-Teller elongation, are complexes containing 2 longer axial and 4 shorter equatorial bonds. Octahedral complexes with Jahn-Teller compression, are the reverse of those with elongation, namely complexes containing 2 shorter axial and 4 longer equatorial bonds instead.

Octahedral Jahn-Teller elongated structures can qualitatively be understood in terms of the d -electron stabilisation energy. If a larger concentration of electron density is found between the metal and the two ligands on the z-axis, than between the metal and the four ligands lying in the xy-plane, namely:

$$e^- \text{ density}(\text{M}-\text{L})_{z\text{-axis}} > e^- \text{ density}(\text{M}-\text{L})_{xy\text{-plane}}$$

then greater electrostatic repulsion will occur between the metal and the two ligands on the z-axis, than with the other four ligands in the xy-plane. The electrostatic repulsion along the z-axis leads to elongation of the bonds along the z-axis (elongation Jahn-Teller distortion). This consequently also results in energy lowering of those d -orbitals by a z factor, namely orbitals d_{z^2} , d_{xz} and d_{yz} , since the bonds along the z-axis are elongated; see Fig. 2(b) for the split of the energy levels under D_{4h} and Fig. 2(c) under D_{2h} symmetry. This lowering of the electronic energy is what drives the distortion. Elongation Jahn-Teller distortion is the most preferred distortion and occurs in most of the cases. Elongation Jahn-Teller is more distinctive when the degeneracy occurs in the

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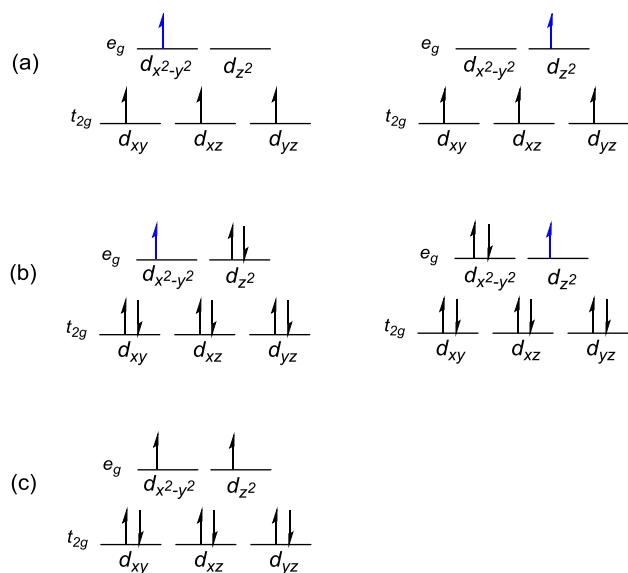


Fig. 1. Octahedral coordination energy level diagrams (e_g and t_{2g} levels), showing the two degenerate states found for the molecular orbitals of the following octahedral transition metal complexes: (a) high spin, $S = 2$, d^4 (as with Cr(II)) and (b) $S = 1/2$, d^9 (as with Cu(II)). (c) High spin, $S = 1$, d^8 octahedral transition metal complexes (such as with Ni(II)) do not have any degenerate states, but only one version of electron filling.

e_g level; for example in high spin d^4 and d^9 octahedral transition metal complexes [2,3]. Conversely, a higher concentration of electron density between the metal and the four ligands in the equatorial plane, than along the z-axis, namely:

$$e^- \text{ density}(\text{M}-\text{L})_{z\text{-axis}} < e^- \text{ density}(\text{M}-\text{L})_{xy\text{-plane}}$$

leads to elongation of bonds along both the x- and y-axes instead, thereby “compressing” the bonds along the z-axis, resulting in compression Jahn-Teller distortion instead.

In order to better understand the role of the higher concentration of electron density between the metal and specific ligands (either on the z-axis or on the x- and y-axes), a computational chemistry experiment utilizing density functional theory (DFT), is described here, for the same complex ion containing three different metals (Cr, Ni, Cu) with different electron filling (d^4 , d^8 , d^9): namely for the high spin, $S = 2$, d^4 hexaaquachromium(II) complex ion, $[\text{Cr}(\text{OH}_2)_6]^{2+}$, the high spin, $S = 1$, d^8 hexaaquanickel(II) complex ion, $[\text{Ni}(\text{OH}_2)_6]^{2+}$, and for the $S = 1/2$, d^9 hexaaquacopper(II) complex ion, $[\text{Cu}(\text{OH}_2)_6]^{2+}$. The results explain the seeming contradiction, namely why an unpaired electron in the d_{z^2} HOMO of a high spin, $S = 2$, d^4 octahedral transition metal compound (leaving $d_{x^2-y^2}$ empty) exhibits elongation (z-out) Jahn-Teller distortion, while an unpaired electron in the same d_{z^2} HOMO of a $S = 1/2$, d^9 octahedral transition metal compound (with a filled electron pair in $d_{x^2-y^2}$) exhibits compression (z-in) Jahn-Teller distortion instead.

2. Computational method

Three complex ions containing different metals (Cr, Ni, Cu), were chosen for this study, namely the high spin $S = 2$, d^4 hexaaquachromium(II) complex ion, $[\text{Cr}(\text{OH}_2)_6]^{2+}$, the $S = 1$, d^8 hexaaquanickel(II) complex ion, $[\text{Ni}(\text{OH}_2)_6]^{2+}$, and the $S = 1/2$, d^9 hexaaquacopper(II) complex ion, $[\text{Cu}(\text{OH}_2)_6]^{2+}$. All three structures were optimized with the density functional theory (DFT) method, Amsterdam Density Functional program (ADF2016) [4]. The B3LYP functional [5,6] and the TZP (Triple ζ polarized) basis set were used. Input geometries without symmetry in the input file, converged either to near T_h symmetry (for $[\text{Ni}(\text{OH}_2)_6]^{2+}$), or to near D_{2h} symmetry (for

both $[\text{Cr}(\text{OH}_2)_6]^{2+}$ and $[\text{Cu}(\text{OH}_2)_6]^{2+}$). The compression and elongated optimized geometries were obtained by submitting compression and elongated geometries as starting geometries respectively. Minima were confirmed by a frequency analysis. The optimized Cartesian coordinates are provided in the Supporting Information. The octahedral restrained models of $S = 2$, d^4 $[\text{Cr}(\text{OH}_2)_6]^{2+}$ and $S = 1/2$, d^9 $[\text{Cu}(\text{OH}_2)_6]^{2+}$ were identified by the minimum on a PES, where only the M–O bonds were restrained and the rest of the molecule was fully optimized. Quantum theory of atoms in molecules (QTAIM) calculations have been performed on these octahedral restrained models, as well as on the T_h restrained symmetry elongation (z-out) geometry of $S = 2$, d^4 $[\text{Cr}(\text{OH}_2)_6]^{2+}$, and on the T_h restrained symmetry compression (z-in) geometry of $S = 1/2$, d^9 $[\text{Cu}(\text{OH}_2)_6]^{2+}$ using Bader’s QTAIM [7,8,9], as implemented in ADF2016 at the same level of theory. The same qualitative picture that is obtained here with the B3LYP functional, is expected with from calculations using other functionals [10].

3. Results and discussion

The B3LYP/TZP relative energies of the different possible spin states for d^8 $[\text{Ni}(\text{OH}_2)_6]^{2+}$, are 0 eV for $S = 1$ and 2.2 eV for $S = 0$. This confirms that the spin state of d^8 $[\text{Ni}(\text{OH}_2)_6]^{2+}$ is $S = 1$, in agreement with experiment [11]. The B3LYP/TZP relative energies of the different possible spin states for d^4 $[\text{Cr}(\text{OH}_2)_6]^{2+}$, are 0 eV for $S = 2$, 1.1 eV for $S = 1$ and 2.3 eV for $S = 0$. The d^4 $[\text{Cr}(\text{OH}_2)_6]^{2+}$ is thereby confirmed to be a $S = 2$ high spin complex, in agreement with experiment [12]. This result also agrees with the fact that from the spectrochemical series, water is generally considered a weak-field ligand, leading to a small energy splitting, especially when coordinated to first row transition metals of low oxidation state (such as Ni and Cr).

The d^9 $[\text{Cu}(\text{OH}_2)_6]^{2+}$ complex has only one spin state, and can only be $S = 1/2$, as is commonly observed for six-coordinate Cu(II) complexes [3]. For both high spin $S = 2$, d^4 $[\text{Cr}(\text{OH}_2)_6]^{2+}$ and $S = 1/2$, d^9 $[\text{Cu}(\text{OH}_2)_6]^{2+}$ complexes, two different electronic states of similar stability (energies) were obtained, one exhibiting elongation and the other compression Jahn-Teller distortion; see data in Table 1. Optimized DFT calculated metal – O bond lengths, for all five different structures of the three selected molecules, are also listed in Table 1.

For the optimized geometry of the $S = 1$, d^8 hexaaquanickel(II) complex ion, $[\text{Ni}(\text{OH}_2)_6]^{2+}$, the nickel – O bond lengths all are at an equal length of 2.078 Å. This calculated bond length agrees well with available experimental $[\text{Ni}(\text{OH}_2)_6]^{2+}$ structures that are of octahedral geometry, with bond lengths that vary between 2.00 and 2.14 Å for the different structures [13]. The octahedral geometry obtained is according to expectation, since the electron filling of the d-based MOs of this high spin $S = 1$, d^8 hexaaquanickel(II) complex has only one possible version, and does not lead to any degenerate electronic states; see Fig. 1 (c).

On the other hand, two different optimized geometries were obtained for the high spin $S = 2$, d^4 hexaaquachromium(II) complex ion, $[\text{Cr}(\text{OH}_2)_6]^{2+}$. In the first geometry, two of the DFT calculated chromium – O bond lengths were longer (by ca. 0.28 Å) than the other four chromium – O bonds. In the other geometry, two of the DFT calculated chromium – O bond lengths were shorter (by ca. 0.18 Å) than the other four chromium – O bonds (see Table 1). This is consistent with both elongation and compression Jahn-Teller distortion being possible for the Cr complex at the spin state $S = 2$, caused by energy lowering of spatially degenerate electronic ground states of the molecular orbitals (MOs).

Similarly, two different optimized geometries also were obtained for the $S = 1/2$, d^9 hexaaquacopper(II) complex ion, $[\text{Cu}(\text{OH}_2)_6]^{2+}$. In both geometries, two of the DFT calculated copper – O bond lengths also were either longer (by ca. 0.27 Å) or shorter (by ca. 0.19 Å) than the other four copper – O bonds, as was also the case with the Cr – O bonds in the chromium complex. This is again consistent with both elongation and compression Jahn-Teller distortion occurring in the two

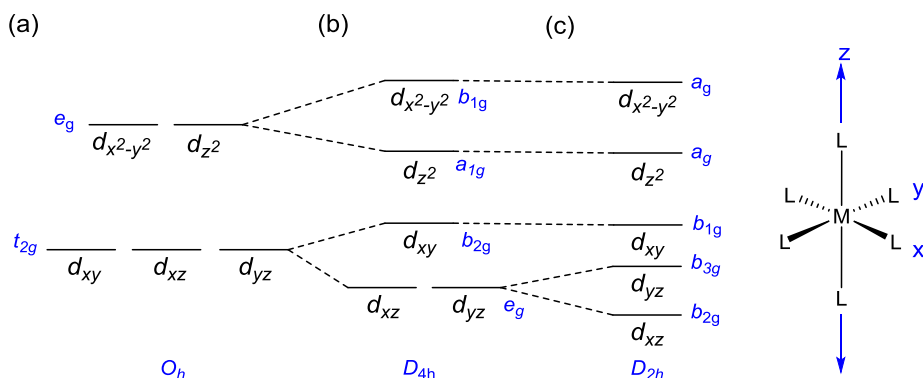


Fig. 2. Schematic energy level diagram of the crystal field splitting in (a) a normal octahedral environment O_h (levels e_g and t_{2g}). Further crystal field splitting occurs, of both the degenerate e_g energy levels, accompanied by the splitting of the t_{2g} level (with all three levels lowered by a z-factor, to a lower energy) under (b) D_{4h} symmetry and under (c) D_{2h} symmetry. This splitting of the energy levels from O_h to both D_{4h} and D_{2h} leads to a Jahn-Teller stabilization energy, accompanied by a tetragonal elongation (z-out) for some complexes with a spatially degenerate electronic ground state, such as high spin $S = 2$, d^4 ($t_{2g}^3 e_g^1$) and $S = 1/2$, d^9 ($t_{2g}^6 e_g^3$) octahedral transition metal complexes. The splittings between energy levels are not drawn to scale.

different optimized structures obtained for Cu (Table 1). The DFT calculated copper – O bond lengths obtained for both the elongation and compression Jahn-Teller distorted optimized structures, fall well into the range of the experimentally observed bond lengths; see the experimental data given in Table 1.

Evaluation of the order of the mainly d -based frontier molecular energy levels of complexes $[\text{Ni}(\text{OH}_2)_6]^{2+}$, $[\text{Cr}(\text{OH}_2)_6]^{2+}$ (with both z-in and z-out geometries) and $[\text{Cu}(\text{OH}_2)_6]^{2+}$ (both z-in and z-out geometries), as well as the relevant orbitals (See Supporting Information, Figs. S2–S3 and Fig. 4), provided the following insights:

- For the high spin $S = 1$, d^8 $[\text{Ni}(\text{OH}_2)_6]^{2+}$ complex, the d -based frontier molecular energy levels, fall into only two groups, namely level e_g (d_{xy} , d_{xz} and d_{yz}) and level t_{2g} (d_{z^2} and $d_{x^2-y^2}$). This arrangement is as predicted for a normal octahedral ligand field, without any degenerate electronic ground states (Fig. S2).
- For the high spin $S = 2$, d^4 $[\text{Cr}(\text{OH}_2)_6]^{2+}$ complex, the LUMO and top four occupied MOs of the optimized geometries of both the elongation (z-out) and compression (z-in) distortions, are Cr- d -based α -MOs of anti-bonding character. In the case of elongation (z-out) distortion of the d^4 complex, these d -based α -frontier molecular energy levels are in the following order of ascending energy: $d_{xy} < d_{xz} < d_{yz} < d_{z^2}$ (HOMO) $< d_{x^2-y^2}$ (LUMO). This order is consistent with the proposed splitting of the degenerate e_g and t_{2g} energy levels of high spin d^4 octahedral transition metal complexes, accompanied by a tetragonal elongation along the z-axis (z-out Jahn-Teller effect) of the d^4 octahedral transition metal complex. Since the LUMO (lowest unoccupied molecular orbital, occupation = 0) is of $d_{x^2-y^2}$ character, and the HOMO (highest occupied molecular orbital, occupation = $1e^-$) of d_{z^2} character, a larger concentration of electron density will occur on the z-axis between the metal and each of the two ligands, than between the metal and each of the four ligands lying in the xy-plane, therefore predicting elongation (z-out) Jahn-Teller distortion for this high spin $S = 2$, d^4

optimized geometry for $[\text{Cr}(\text{OH}_2)_6]^{2+}$; see Fig. 3(a) and (c) and Fig. S3(a).

- In the case of compression (z-in) distortion of the same d^4 $[\text{Cr}(\text{OH}_2)_6]^{2+}$ complex, the d -based α -frontier molecular energy levels are in the following order of ascending energy: $d_{xz} < d_{yz} < d_{xy} < d_{x^2-y^2}$ (HOMO) $< d_{z^2}$ (LUMO). This order is consistent with the proposed splitting of the degenerate e_g and t_{2g} energy levels of high spin d^4 octahedral transition metal complexes, accompanied by a tetragonal compression along the z-axis (z-in Jahn-Teller effect). Since the LUMO (occupation = 0) is of d_{z^2} character, and the HOMO (occupation = $1e^-$) of $d_{x^2-y^2}$ character, a larger concentration of electron density will occur between the metal and each of the four ligands lying in the xy-plane, than between the metal and each of the two ligands along the z-axis, therefore predicting compression (z-in) Jahn-Teller distortion for this high spin $S = 2$, d^4 geometry for $[\text{Cr}(\text{OH}_2)_6]^{2+}$; see Fig. 3(d) and (f) and Fig. S3(b).
- On the other hand, for the $S = 1/2$, d^9 $[\text{Cu}(\text{OH}_2)_6]^{2+}$ complex, the HOMO and the LUMO of the optimized geometries of both elongation (z-out) and compression (z-in) distortions are both β -orbitals, contrary to the above α -MOs. The d -occupation of the elongation (z-out) geometry is $d_{xy}^2 d_{xz}^2 d_{yz}^2 d_{z^2}^1 d_{x^2-y^2}^1$, i.e. the SOMO (singly occupied molecular orbital) and LUMO both are of $d_{x^2-y^2}$ character. Thus, since for the elongation (z-out) $S = 1/2$, d^9 $[\text{Cu}(\text{OH}_2)_6]^{2+}$ geometry, both the α - d_{z^2} and β - d_{z^2} orbitals are occupied (occupation = $2e^-$), but only the α - $d_{x^2-y^2}$ orbital is occupied (occupation = $1e^-$, since β - $d_{x^2-y^2}$ is empty), a larger concentration of electron density will occur on the z-axis between the metal and the two ligands (d_{z^2} with occupation = $2e^-$), than between the metal and the four ligands lying in the xy-plane ($d_{x^2-y^2}$ with occupation = $1e^-$), therefore predicting elongation (z-out) Jahn-Teller distortion for this $S = 1/2$, d^9 $[\text{Cu}(\text{OH}_2)_6]^{2+}$ geometry, see Fig. 3(b) and (c).
- In the case of compression (z-in) distortion of the same d^9 complex, the d -occupation of the compression (z-in) $[\text{Cu}(\text{OH}_2)_6]^{2+}$

Table 1

DFT relative energies (eV) and average metal–O bond lengths (Å), as calculated for the indicated complexes, with the range of experimentally obtained bond length values from X-ray data, given in brackets as a comparison.

Complex	d electrons	Spin state	ΔE (eV)	M – L (1) (z-axis ave) in Å	M – L (2) (xy-plane ave) in Å	difference (1)–(2) in Å	Jahn-Teller effect
$[\text{Ni}(\text{OH}_2)_6]^{2+}$	d^8	$S = 1$	–	2.078 (2.00–2.14) ^a	2.078 (2.00–2.14) ^a	0.00	no elongation
$[\text{Cr}(\text{OH}_2)_6]^{2+}$	d^4	$S = 2$	0.00	2.384 ^b	2.104 ^b	0.28	z-out elongation
			0.03	2.072 ^b	2.247 ^b	–0.18	z-in compression ^c
$[\text{Cu}(\text{OH}_2)_6]^{2+}$	d^9	$S = 1/2$	0.00	2.289 (2.16–2.64) ^d	2.024 (1.81–2.09) ^d	0.27	z-out elongation
			0.02	1.980 (1.95–1.99) ^e	2.169 (2.00–2.36) ^e	–0.19	z-in compression

^a Average value from 270 crystals obtained from the CSD [13].

^b No experimental X-ray data available on CSD [13].

^c Have one imaginary frequency of 0 strength.

^d Average value from 45 crystals showing z-out elongation, from CSD [13].

^e Average value from 5 crystals showing z-in compression, from CSD [13].

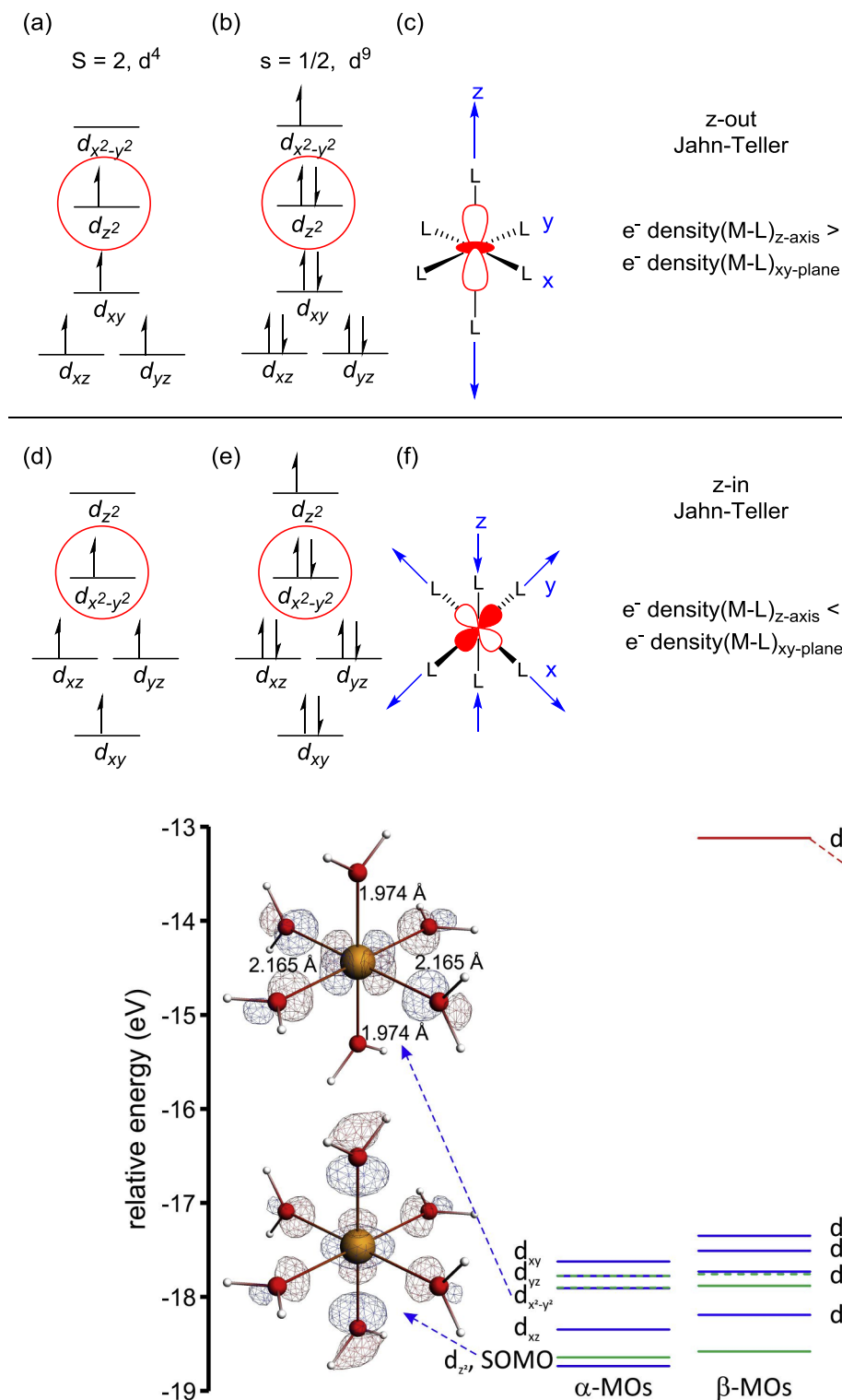


Fig. 3. Top: Schematic energy level diagram of the preferred crystal field splitting in the following octahedral transition metal complexes, which show a larger concentration of electron density between the metal and the two ligands on the z-axis, than between the metal and the four ligands lying in the xy-plane, namely: (a) the high spin $S = 2$, d^4 octahedral complex (such as with Cr(II)), and (b) the $S = 1/2$, d^9 complex (such as with Cu(II)). (c) This larger concentration of electron density along the z-axis, leads to elongation (z-out) Jahn-Teller distortion. Bottom: Schematic energy level diagram of the crystal field splitting (the alternative less observed option) in the following complexes, which show a lower concentration of electron density between the metal and the two ligands on the z-axis, than between the metal and the four ligands lying in the xy-plane, namely: (d) the high spin $S = 2$, d^4 and (e) $S = 1/2$, d^9 octahedral transition metal complexes. (f) This lower concentration of electron density along the z-axis, leads to compression (z-in) Jahn-Teller distortion.

Fig. 4. Molecular orbital (MO) energy level diagram, of the B3LYP compression (z-in) Jahn-Teller geometry, of the $S = 1/2$, d^9 hexaaquacopper(II) complex ion, $[\text{Cu}(\text{OH}_2)_6]^{2+}$. Selected frontier MO levels, selected MOs (α-MO and β-MO) and bond lengths (Å) are shown. Blue lines indicate the energy levels of filled d-based MOs, while red lines indicate the empty d-based LUMO. Filled MOs without significant metal-d character are shown in green. A T_h restrained geometry was used to construct the diagram.

geometry is: $d_{xy}^2 d_{xz}^2 d_{yz}^2 d_{x^2-y^2}^2 d_{z^2}^1$, i.e. the SOMO and LUMO both are of d_{z^2} character, see Fig. 4. For this compression (z-in) geometry, both the α and β- $d_{x^2-y^2}$ orbitals are occupied (occupation = $2e^-$), the d_{z^2} orbital is singly occupied (SOMO, occupation = $1e^-$) and β- d_{z^2} is the LUMO (occupation = 0). In this case a larger concentration of electron density will occur between the

metal and the four ligands lying in the xy-plane ($d_{x^2-y^2}$ is doubly occupied, occupation = $2e^-$), than between the metal and the two ligands on the z-axis (d_{z^2} with occupation = $1e^-$), therefore predicting compression (z-in) Jahn-Teller distortion for this $S = 1/2$, d^9 $[\text{Cu}(\text{OH}_2)_6]^{2+}$ geometry; see the bond lengths indicated in Fig. 4.

- (vi). An unpaired electron in the d_{z^2} orbital of a high spin $S = 2$, d^4 octahedral transition metal compound (such as with Cr(II)) exhibits elongation (z-out) Jahn-Teller distortion, while an unpaired electron in the $d_{x^2-y^2}$ orbital of a $S = 1/2$, d^9 octahedral transition metal compound (such as with Cu(II)), also exhibits elongation Jahn-Teller distortion; see Fig. 3(a)–(c).
- (vii). On the other hand, if the unpaired electron instead is in the d_{z^2} orbital of a $S = 1/2$, d^9 octahedral transition metal compound, compression (z-in) Jahn-Teller distortion (Fig. 3 (e) and (f)) will occur instead, while an unpaired electron in the $d_{x^2-y^2}$ orbital of a $S = 1/2$, d^9 octahedral transition metal compound exhibits elongation (z-out) Jahn-Teller distortion instead; see Fig. 3(b) and (c).
- (viii). For molecules containing degenerate molecular energy levels, the exact order of the splitted energy levels (e_g and t_{2g} levels) is not always exactly as predicted by Figs. 2 and 3, however it does follow the general trend as predicted. The exact order depends on the electron spin directions of the electrons (α or β), as well as on the electronic environment caused by the ligands.

In order to find computational support for the physical origin of the observed Jahn-Teller distortions, as ascribed by electrostatics due to differences in electron density (Fig. 3), quantum theory of atoms in molecules (QTAIM) calculations have been performed on octahedrally restrained models, as well as on the T_h restrained elongation (z-out) geometry of $S = 2$, d^4 $[\text{Cr}(\text{OH}_2)_6]^{2+}$, and on the T_h restrained compression (z-in) geometry of $S = 1/2$, d^9 $[\text{Cu}(\text{OH}_2)_6]^{2+}$. The octahedrally restrained model was identified by the minimum on a PES, where only the M–O bonds were restrained and the rest of the molecule was fully optimized. QTAIM analysis of a molecular system uses the electron density of molecules as a tool to study the nature of bonding in molecular systems. Bonds between atoms are identified by bond paths (BP) between atoms (atom critical points ACP), with bond critical points (BCP) [14]. The QTAIM theory provides topological descriptors, such as the electron density ($\rho/\text{e a}_0^{-3}$) and the Laplacian of electron density ($\nabla^2\rho/\text{e a}_0^{-5}$) at the BCP, which have been used to characterize the strength of bonds in various molecular systems [15].

Fig. 5 visualizes the results obtained for $S = 1/2$, d^9 $[\text{Cu}(\text{OH}_2)_6]^{2+}$, both for the octahedrally restrained model (Fig. 5(a)), as well as for the optimized compression (z-in) geometry (Fig. 5(b)). The electron occupation of both molecules is $d_{xy}^1 d_{yz}^1 d_{zx}^1 d_{x^2-y^2}^2 d_{z^2}^1$, i.e. both the α and β - $d_{x^2-y^2}$ orbitals are occupied (occupation = $2e^-$), while the d_{z^2} orbital is singly occupied (SOMO, occupation = $1e^-$), with the octahedrally restrained model 0.16 eV higher in energy than the compression (z-in) geometry on the right. The QTAIM determined electron density ρ (0.054 e a_0^{-3}) at the BCP of the Cu–O bonds along the z-axis of the octahedrally restrained model, is less than the electron density ρ (0.058 e a_0^{-3}) at the BCP of the Cu–O bonds in the xy-plane. This result, as well as evaluation of the DFT calculated electron density isosurface of the molecule, show that:

$$e^- \text{ density}(\text{Cu-L})_{z\text{-axis}} < e^- \text{ density}(\text{Cu-L})_{xy\text{-plane}} \text{ (Fig. 5(a))}$$

In this case, the larger concentration of electron density is found between the metal and the four ligands lying in the xy-plane of the octahedrally restrained model, which exhibits degenerate molecular orbital energies. This electron density in the xy-plane will distort itself in such a way that the degenerate energy levels will split, resulting in the lowering of symmetry of the molecule, as well as in the lowering of the overall energy of the molecule. This results in the compression (z-in) Jahn-Teller geometry of $S = 1/2$, d^9 $[\text{Cu}(\text{OH}_2)_6]^{2+}$; as the shorter bond lengths along the z-axis indicate in Fig. 5(b).

Fig. 6 visualizes the results obtained for high spin $S = 2$, d^4 $[\text{Cr}(\text{OH}_2)_6]^{2+}$, both for the octahedrally restrained model (Fig. 6(a)), as well as for the T_h restrained elongation (z-out, $E = 0 \text{ eV}$) geometry (Fig. 6(b)). The electron occupation of both molecules is $d_{xy}^1 d_{yz}^1 d_{zx}^1 d_{x^2-y^2}^0 d_{z^2}^2$, therefore the d_{z^2} orbital is singly occupied (HOMO, occupation = $1e^-$), with the octahedrally restrained model 0.17 eV higher in energy than the elongation (z-out) geometry on the right. In this case, the QTAIM determined electron density of the octahedrally restrained model ρ (0.053 e a_0^{-3}) at the BCP of the Cu–O bonds along the z-axis, is larger than the electron density ρ (0.045 e a_0^{-3}) at the BCP of the Cu–O bonds in the xy-plane. This result, as well as evaluation of the DFT calculated electron density isosurface of the molecule show that:

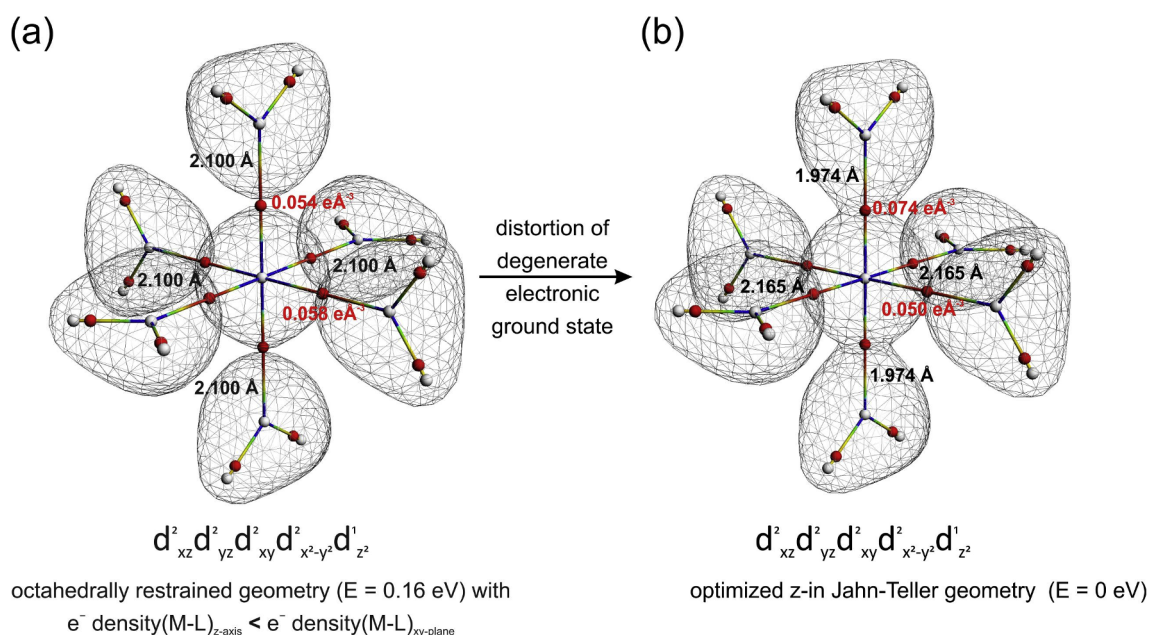


Fig. 5. Z-in compression for the d^9 transition metal complex: Visualization of the QTAIM determined intramolecular bond-paths (BP), bond critical points (BCP in red) and atom critical points (ACP in white), for (a) the octahedrally restrained model (T_h symmetry, left), and for (b) the T_h restrained compression Jahn-Teller (z-in, right) geometry of the complex $S = 1/2$, d^9 $[\text{Cu}(\text{OH}_2)_6]^{2+}$. Selected bond lengths (Å) are shown in black font, and QTAIM determined electron density ρ (e a_0^{-3}) at the BCP, in red font. The colour of the bond-paths changes according to electron density, from blue (high density) to green to red (low density). The B3LYP calculated electron density isosurface of the molecule is shown in grey, using a contour of 0.057 e a_0^{-3} .

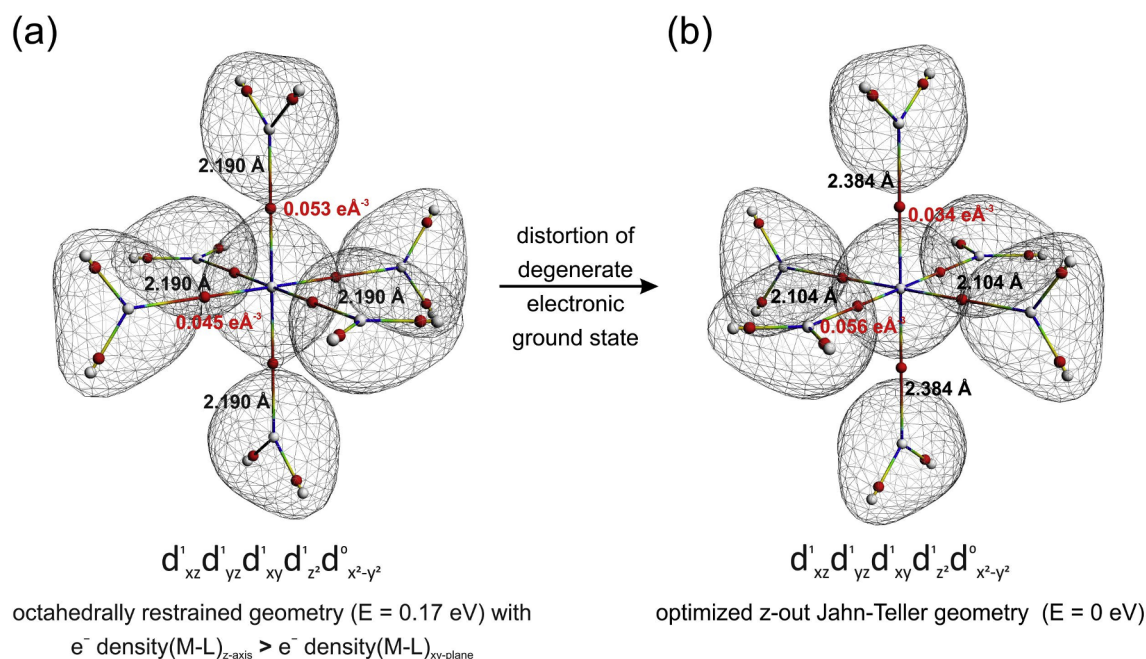


Fig. 6. Z-out elongation for the d^4 transition metal complex: Visualization of the QTAIM determined intramolecular bond-paths (BP), bond critical points (BCP in red) and atom critical points (ACP in white), for (a) the octahedrally restrained model (T_h symmetry, left), and for (b) the T_h restrained elongation Jahn-Teller (z-out, right) geometry of the complex $S = 2$, d^4 $[\text{Cr}(\text{OH}_2)_6]^{2+}$. Selected bond lengths (Å) are shown in black font, and QTAIM determined electron density ρ (e a_0^{-3}) at the BCP, in red font. The colour of the bond-paths changes according to electron density, from blue (high density) to green to red (low density). The B3LYP calculated electron density isosurface of the molecule is shown in grey, using a contour of 0.052 e/a_0^3 .

e^- density(Cu-L)_{z-axis} > e^- density(Cu-L)_{xy-plane} (Fig. 6(a))

The result of this higher density, is greater electrostatic repulsion between Cu and the two ligands on the z-axis, than between Cu and the other four ligands in the xy-plane, which leads to elongation of the bonds along the z-axis (elongation Jahn-Teller distortion) and a lowering in energy for z-out elongation of $S = 2$, d^4 $[\text{Cr}(\text{OH}_2)_6]^{2+}$; as the longer bond lengths along the z-axis indicate in Fig. 6 (b) on the right.

QTAIM analysis of the optimized geometry of the $S = 1$ d^8 hexa-aquanickel(II) complex ion, $[\text{Ni}(\text{OH}_2)_6]^{2+}$, showed that the electron density ρ at the BCP of all the Ni-O bonds are the same, namely 0.057 e a_0^{-3} , thus

e^- density(Ni-L)_{z-axis} = e^- density(Ni-L)_{xy-plane}

Therefore there is no imbalanced electrostatic repulsion between Ni and any of the ligands on the z-, x- or y-axis, thus no distortion and consequently $[\text{Ni}(\text{OH}_2)_6]^{2+}$ exhibits an octahedral geometry.

4. Conclusions

The Jahn-Teller effect, which is the geometrical effect on a molecule, due to the energy lowering of spatially degenerate electronic ground states, is not determined, as is commonly perceived, by the shape of the highest occupied molecular orbital (HOMO), but determined rather by the difference in the electron density concentration (before distortion) between the metal and the two ligands on the z-axis, and the electron density concentration between the metal and the four ligands lying in the xy-plane. If this electron density concentration is higher along the z-axis, this imbalance in electron density will lead to elongation (z-out) Jahn-Teller distortion. On the other hand, if the electron density is lower along the z-axis, this imbalance will lead to compression (z-in) Jahn-Teller distortion instead. Accordingly, high spin $S = 2$, d^4 octahedral metal-complexes with a d_{z^2} HOMO containing an unpaired electron, exhibit elongation (z-out) Jahn-Teller distortion (since $d_{x^2-y^2}$ is empty in comparison); while $S = 1/2$, d^9 octahedral metal-complexes, also with a d_{z^2} HOMO containing an unpaired

electron, exhibit compression (z-in) Jahn-Teller distortion instead (since $d_{x^2-y^2}$ contains an electron pair in comparison).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jica.2018.10.040>.

References

- [1] H.A. Jahn, E. Teller, Stability of polyatomic molecules in degenerate electronic states. i. orbital degeneracy, P. R. Soc. London, 1934–1990. Series A, Mathemat. Phys. Sci. 161 (905) (1937) 220–235 <http://www.jstor.org/stable/96911>.
- [2] R. Freitag, J. Conradie, Understanding the Jahn-Teller Effect in octahedral transition metal complexes – a Molecular Orbital view of the $\text{Mn}(\beta\text{-diketonato})_3$ complex, J. Chem. Educat. 90 (12) (2012) 1692–1696, <https://doi.org/10.1021/ed400370p>.
- [3] M.A. Halcrow, Jahn-Teller distortions in transition metal compounds, and their importance in functional molecular and inorganic materials, Chem. Soc. Rev. 42 (4) (2013) 1784–1795, <https://doi.org/10.1039/c2cs35253b>.
- [4] F.M. Bickelhaupt, E.J. Baerends, C.F. Guerra, S.J.A. van Gisbergen, J.G. Snijders, T.J. Ziegler, The ADF program system was obtained from Scientific Computing and Modeling, Amsterdam (<http://www.scm.com/>). For a description of the methods used in ADF, see: ee: G. te Velde, F.M. Bickelhaupt, E.J. Baerends, C.F. Guerra, S.J.A. van Gisbergen, J.G. Snijders, T.J. Ziegler Chemistry with ADF, J. Computat. Chem. 22 (2001) 931–967 [10.1002/jcc.1056](https://doi.org/10.1002/jcc.1056).
- [5] A.D. Becke, Density-functional thermochemistry. III. The role of exact exchange, J. Chem. Phys. 98 (1993) 5648–5652, <https://doi.org/10.1063/1.464913>.
- [6] C. Lee, W. Yang, R.G. Parr, Development of the Colle-Salvetti correlation-energy formula into a functional of the electron density, Phys. Rev. B 37 (1988) 785–789,

- <https://doi.org/10.1103/PhysRevB.37.785>.
- [7] R.F.W. Bader, A quantum theory of molecular structure and its applications, *Chem. Rev.* 91 (1991) 893–928, <https://doi.org/10.1021/cr00005a013>.
- [8] F. Cortés-Guzmán, R.F.W. Bader, Complementarity of QTAIM and MO theory in the study of bonding in donor–acceptor complexes, *Coordination Chem. Rev.* 249 (2005) 633, <https://doi.org/10.1016/j.ccr.2004.08.022>.
- [9] J.I. Rodríguez, R.F.W. Bader, P.W. Ayers, C. Michel, A.W. Götz, C. Bo, A high performance grid-based algorithm for computing QTAIM properties, *Chem. Phys. Lett.* 472 (2009) 149–152, <https://doi.org/10.1016/j.cplett.2009.02.081>.
- [10] M. Bühl, H. Kabrede, Geometries of transition-metal complexes from density-functional theory, *J. Chem. Theory Comput.* 2 (2006) 1282–1290, <https://doi.org/10.1021/ct6001187>.
- [11] (a) J.H.E. Griffiths, J. Owen, Paramagnetic Resonance in the Nickel Tutton Salts *Proceed. R. Soc. London. Ser. A, Mathemat. Phys. Sci.* 123: 1115 1952 459 473 <https://www.jstor.org/stable/99119>.
(b) N.W. Grimes, M.W. Webb The Symmetry of the Crystalline Field of the $(\text{Ni}(\text{H}_2\text{O})_6)^{2+}$ Complex *Proceed. Phys. Soc.* 82 (6) 10.1088/0370-<http://iopscience.iop.org/article/10.1088/0370-1328/82/6/134/meta>.
(c) A. Bose, R. Chatterjee, Orthorhombic Crystalline Field Theory of $\text{Ni}^{2+} \cdot 6\text{H}_2\text{O}$ Complex, *Proceedings of the Physical Society.* 82 23 (1963) 23-. <http://iopscience.iop.org/article/10.1088/0370-1328/82/1/305>.
- [12] J. Telser, L.A. Pardi, J. Krzystek, L.-C. Brunel, Inorganic EPR Spectra from “EPR-Silent” Species: high-Field EPR Spectroscopy of Aqueous Chromium(II), *Chem.* 37 (22) (1998) 5769–5775, <https://doi.org/10.1021/ic9806683>.
- [13] Cambridge Structural Database (CSD), Version 5.39, update Feb 2018 Cambridge UK 2017.
- [14] R.F.W. Bader, A bond path: a universal indicator of bonded interactions, *J. Phys. Chem. A* 102 (1998) 7314–7323, <https://doi.org/10.1021/jp981794v>.
- [15] P.S.V. Kumar, V. Raghavendra, V. Subramanian, Bader's theory of atoms in molecules (AIM) and its applications to chemical bonding, *J. Chem. Sci.* 128 (2016) 1527–1536, <https://doi.org/10.1007/s12039-016-1172-3>.